KUTAISI INTERNATIONAL UNIVERSITY

Scientific Computing

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2.3: geometry of Euler's method in 11 examples

Outline

Lecture 2.3: geometry of Euler's method in 11 examples

- Slope field and tangent steps
- Growth, decay, forced and logistic equations
- Step-size study: first order
- Oscillators, closed orbits and stiffness
- Q & A

Series: 2.1 Euler schemes. 2.2 errors. 2.3 geometry. 2.4 explicit convergence. 2.5 Grönwall. 2.6 implicit convergence.

Remark: geometric meaning of an Euler step

Remark 2.1 (Euler follows tangents)

- The ODE $y' = f(t, y)$ prescribes a **slope** at every point (t, y): the slope (direction) field.

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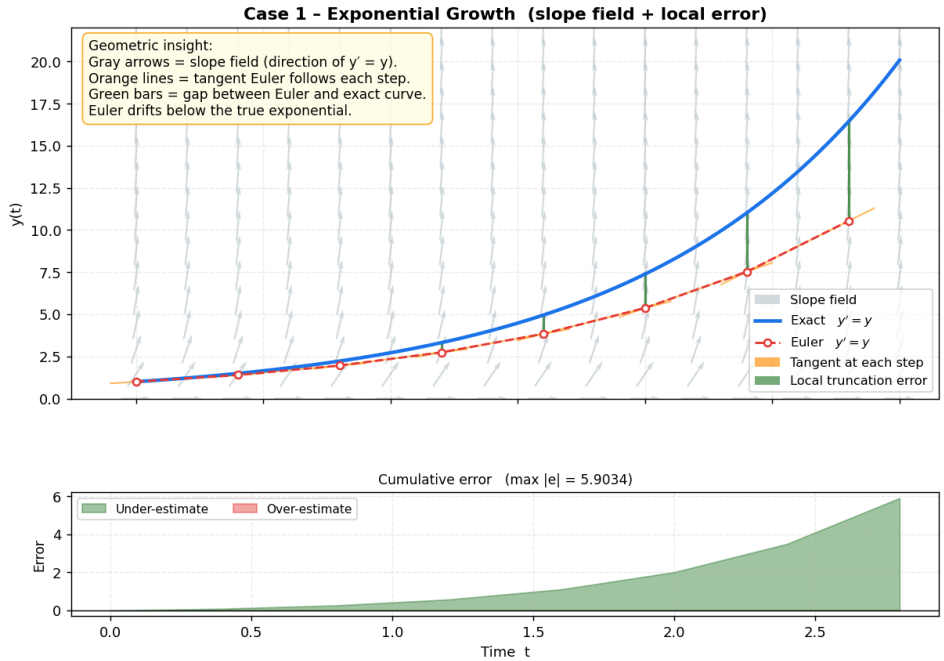
- The ODE $y' = f(t, y)$ prescribes a **slope** at every point (t, y): the slope (direction) field.
- A solution is a curve tangent to the field everywhere.
- An Euler step from (t_n, y_n) moves along the tangent line with slope $f(t_n, y_n)$ for a time h; the end point is (t_{n+1}, y_{n+1}).

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- The curve bends away from its tangent; the gap after one step is the LTE, the accumulated gap is the GTE.

Figure: slope field of $y' = y$

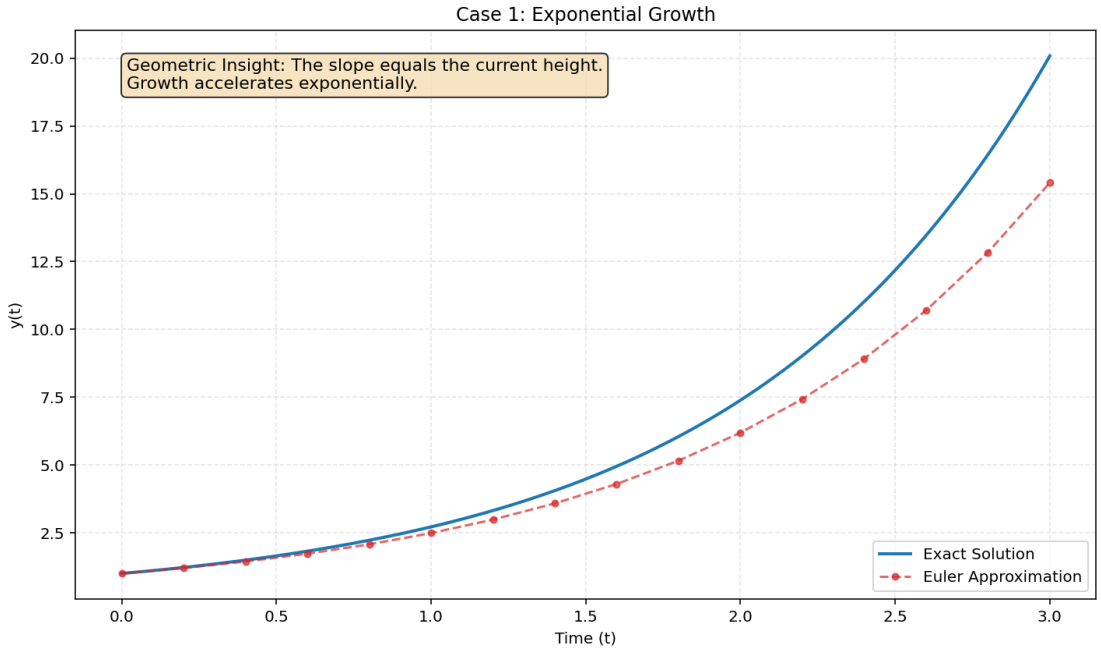


Gray arrows: slope of the solution through each point. Solid blue: exact solution $y = e^t$.

How to read this figure

The arrows give the direction that $y' = y$ dictates; the blue curve is tangent to every arrow it meets. Euler builds its path from these arrows.

Figure: exponential growth $y' = y$

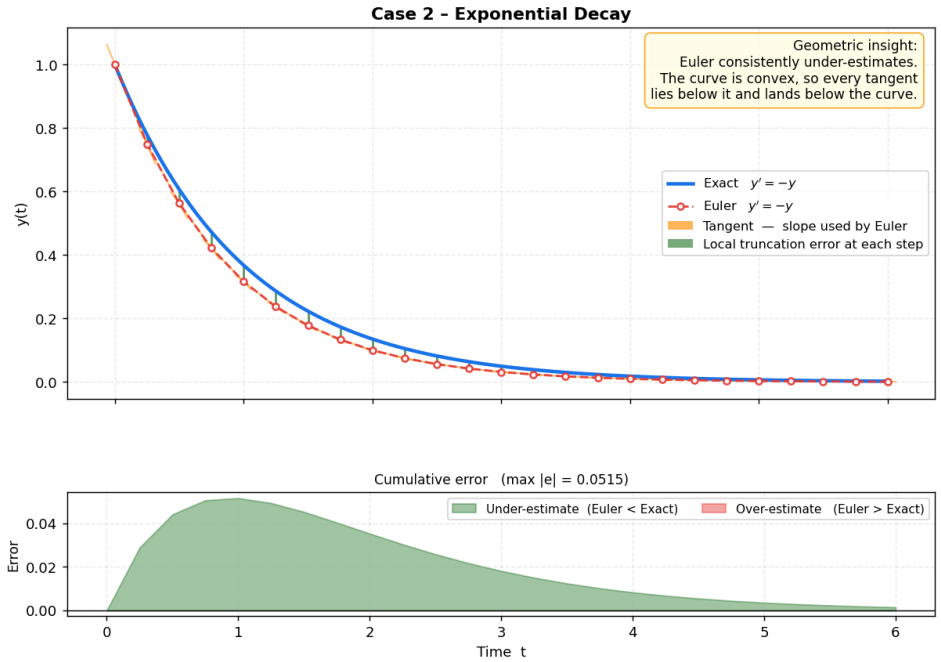


Exact $y = e^t$ (solid blue), Euler path (dashed red), Euler tangents (orange).

How to read this figure

Orange segments are the tangents Euler follows at each step; green vertical bars are the LTE of each step; the bottom panel is the cumulative error (green fill: Euler under-estimates, red fill: over-estimates). Euler drifts below the true exponential because the solution is convex.

Figure: exponential decay $y' = -y$

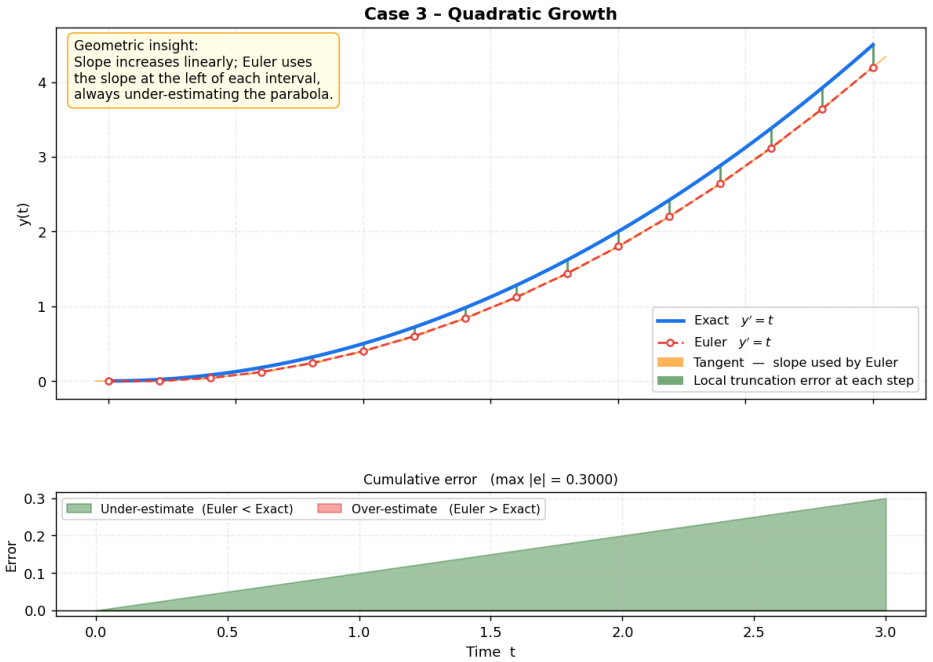


Exact $y = e^{-t}$ (solid blue), Euler steps (dashed red).

How to read this figure

The solution e^{-t} is convex, so each orange tangent lies below the curve. Euler lands below the true curve and under-estimates (green fill in the bottom panel).

Figure: quadratic growth $y' = t$

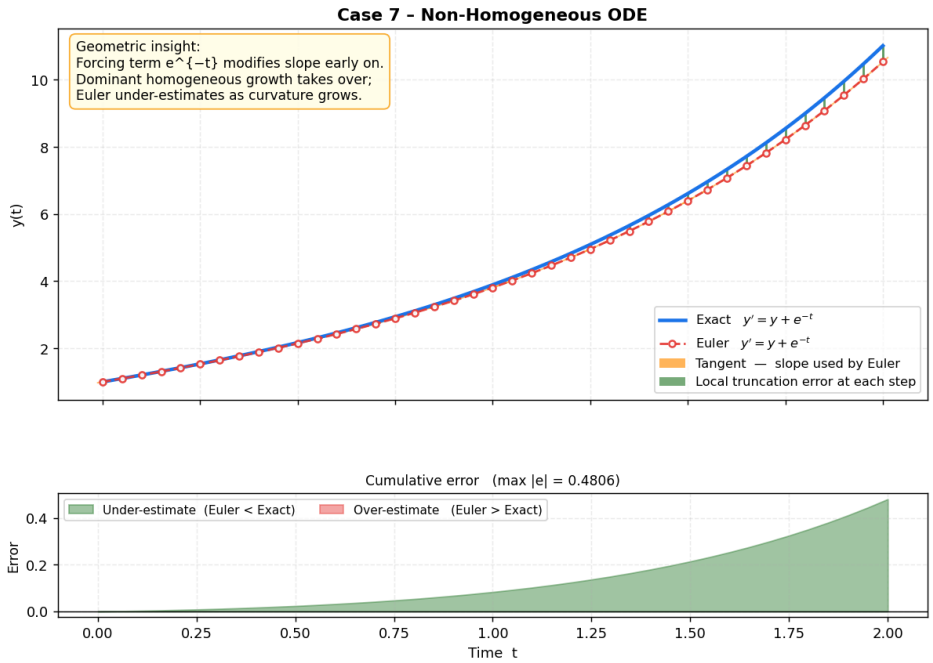


Exact $y = \frac{1}{2}t^2$ (solid blue), Euler path (dashed red).

How to read this figure

The slope grows linearly, but Euler uses the slope at the left end of each interval, so it always trails the curve and under-estimates (green bars and green fill).

Figure: non-homogeneous ODE $y' = y + e^{-t}$

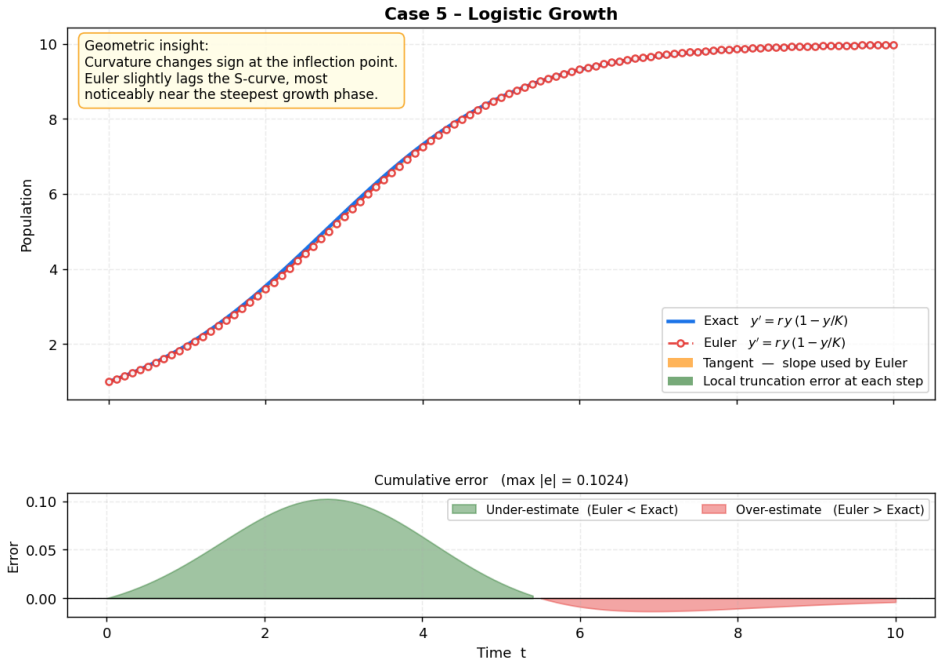


Exact $y = \frac{3}{2}e^{t} - \frac{1}{2}e^{-t}$ (solid blue), Euler (dashed red).

How to read this figure

The forcing e^{-t} bends the slope early on, then the growing homogeneous part dominates. Euler under-estimates as the curvature increases.

Figure: logistic growth $y' = ry(1 - y/K)$

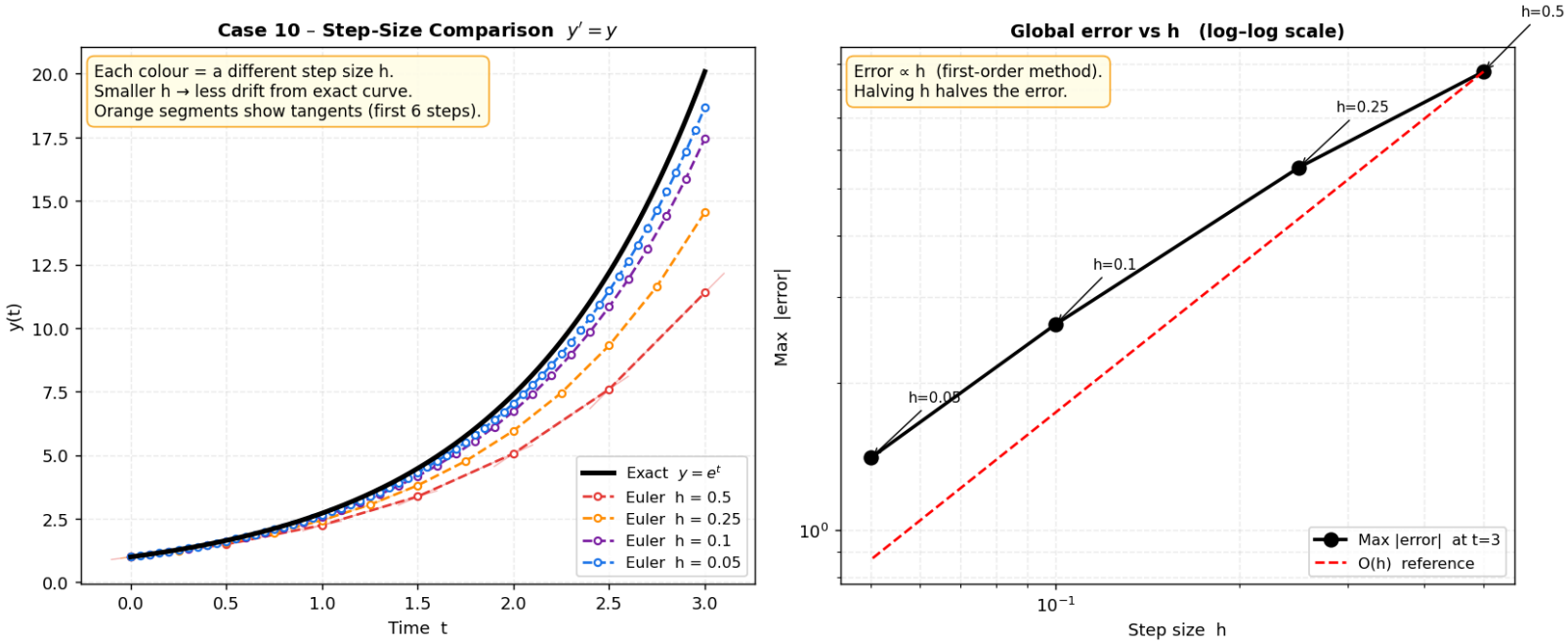


S-shaped solution approaching the carrying capacity K (solid blue), Euler (dashed red).

How to read this figure

The slope changes with y; Euler lags most around the inflection point, where the curvature is largest. The bottom panel shows the cumulative error.

Figure: effect of the step size on $y' = y$

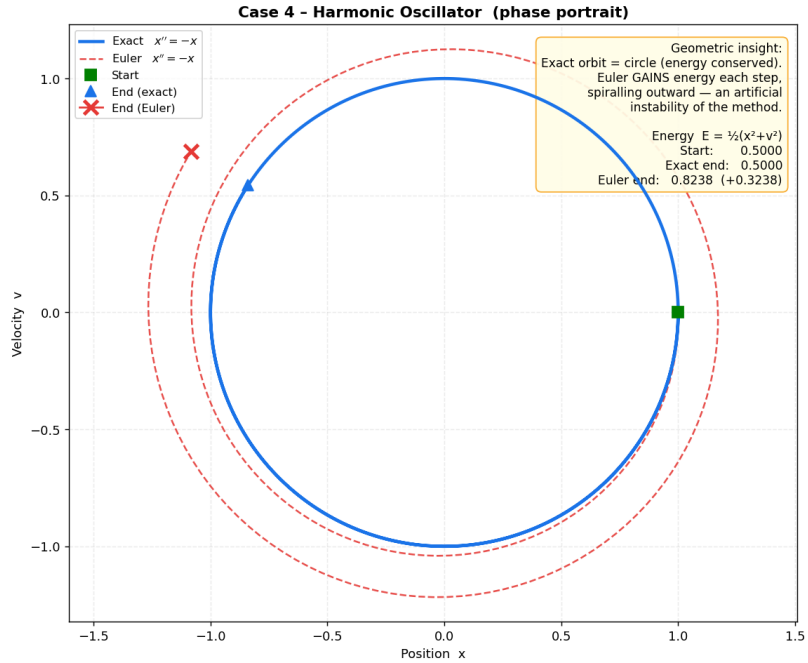


Left: four Euler solutions ($h = 0.5, 0.25, 0.1, 0.05$) and the exact e^t. Right: global error versus h, log–log scale.

How to read this figure

Smaller h stays closer to the exact curve. On the log–log plot the points lie on a line of slope 1: the error is proportional to h (first order), so halving h halves the error.

Figure: harmonic oscillator $x'' = -x$

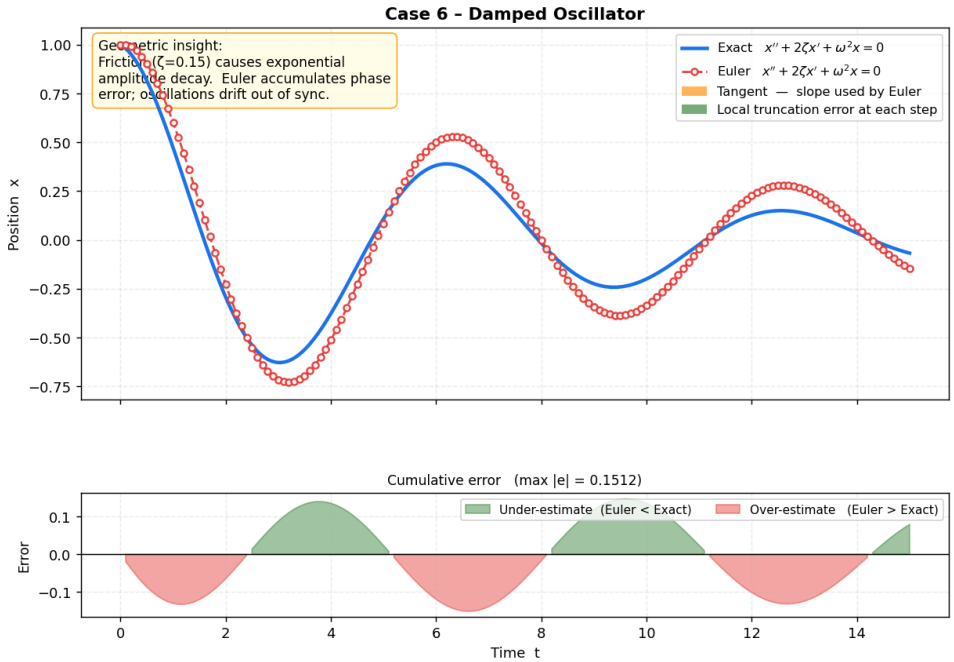


Phase portrait: exact circular orbit (solid blue) versus Euler's spiral (dashed red).

How to read this figure

Horizontal axis: position x; vertical axis: velocity $\dot{x}$. The exact orbit is a closed circle because the energy $E = \frac{1}{2}(x^2 + v^2)$ is conserved. Euler gains energy at every step and spirals outward: an artificial instability of the method.

Figure: damped oscillator

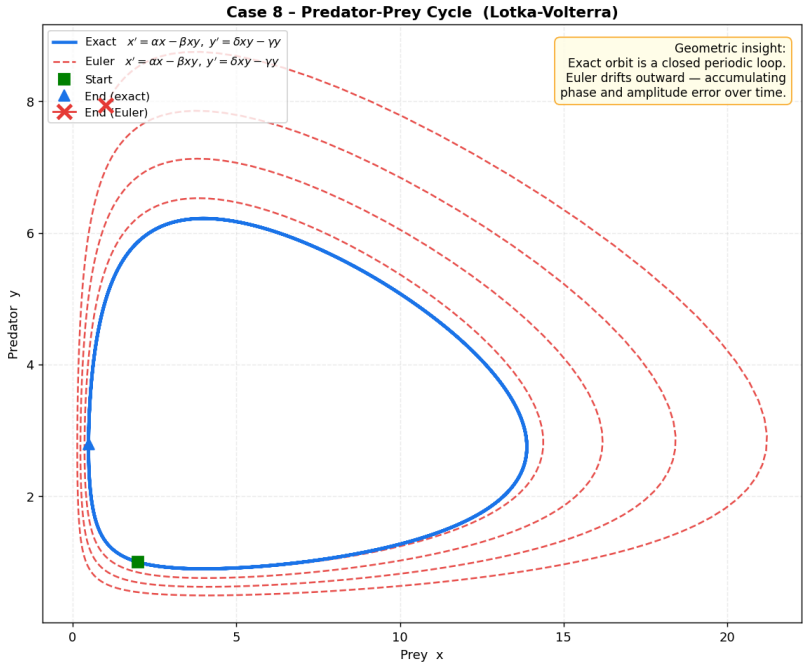


$x'' + 2\zeta x' + \omega^2 x = 0$, $\zeta = 0.15$: exact decaying oscillation (solid blue), Euler (dashed red).

How to read this figure

Friction produces the decaying envelope $e^{-\zeta t}$. Euler accumulates a phase error, so its oscillations drift out of step with the exact solution.

Figure: predator–prey (Lotka–Volterra)

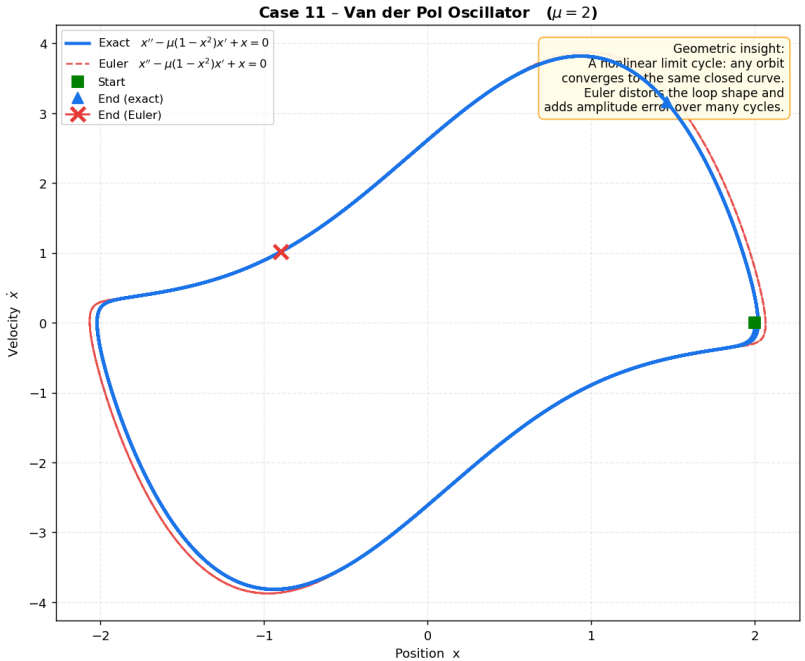


Plane (prey x, predator y): exact closed orbit (solid blue), Euler (dashed red).

How to read this figure

The true system returns to its starting point every cycle. Euler accumulates phase and amplitude error and spirals away from the loop.

Figure: Van der Pol oscillator

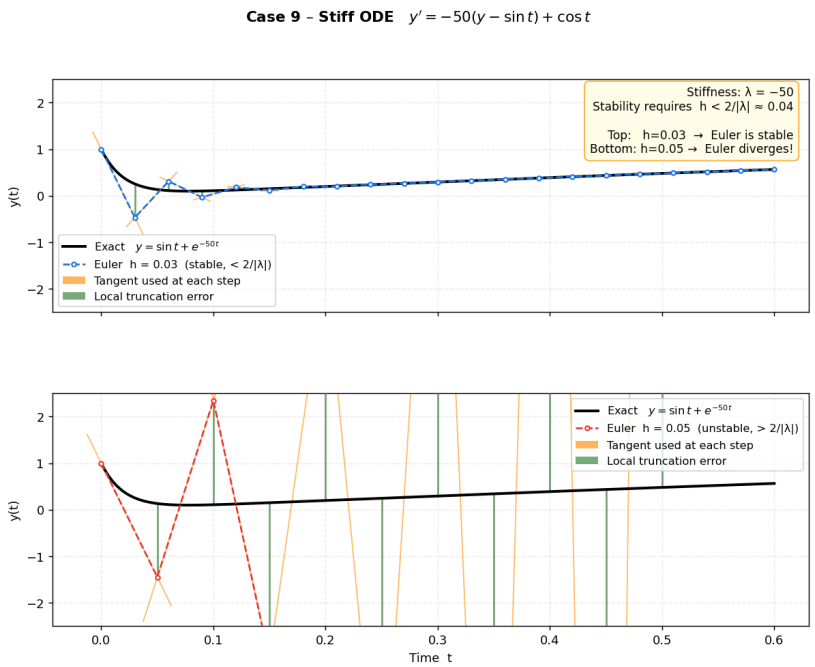


$x'' - \mu(1 - x^2)x' + x = 0$, $\mu = 2$: exact limit cycle (solid blue), Euler (dashed red).

How to read this figure

Phase portrait, position x horizontal, velocity $\dot{x}$ vertical. Green S: start; blue triangle: exact end; red x: Euler end. The Euler spiral distorts the closed limit cycle and grows in amplitude.

Figure: stiff ODE and the step-size limit



$y' = -50(y - \sin t) + \cos t$: Euler with $h = 0.03$ (top) and $h = 0.05$ (bottom), exact solution in black.

How to read this figure

With $\lambda = -50$, stability of explicit Euler needs $h < 2/|\lambda| = 0.04$. The smaller step stays bounded; the larger one oscillates and blows up.

Remark: what the gallery shows

Remark 2.2 (Three lessons from the figures)

- **Sign of the error:** Euler under-estimates convex solutions ($y' = y$, $y' = -y$, $y' = t$): the tangents lie below the curve.

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- **Sign of the error:** Euler under-estimates convex solutions ($y' = y$, $y' = -y$, $y' = t$): the tangents lie below the curve.
- **Accuracy:** the error is proportional to h (first order).

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- **Sign of the error:** Euler under-estimates convex solutions ($y' = y$, $y' = -y$, $y' = t$): the tangents lie below the curve.
- **Accuracy:** the error is proportional to h (first order).
- **Qualitative failure:** on oscillators and closed orbits Euler adds energy and spirals out; on stiff problems it needs $h < 2/|\lambda|$.

Q & A